SIBUSISO MPHO MPOFU

Mechatronics and Industrial Instrumentation Engineering Graduate

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PROFESSIONAL PROFILE

Engineering graduate with ~6-months of industrial control and instrumentation experience at Morupule B Power Station and practical experience supporting oxygen plant installation and commissioning. Background in instrumentation calibration, fault isolation, loop verification, embedded systems, condition monitoring, Python-based analysis and technical documentation. Award-winning final-year project contributor with experience integrating hardware and software systems.

TECHNICAL SKILLS

Instrumentation and maintenance: RTDs, thermocouples, pressure transmitters, motor-operated valves (MOVs), calibration, loop verification, fault diagnosis, preventive and corrective maintenance, lockout/tagout (LOTO)

Embedded systems and electronics: ESP32, STM32 development tools (CubeMX, CubeIDE, CubeProgrammer), Arduino, Jetson Nano, UART, SPI, I2C, PWM, basic Modbus, Proteus, Multisim

Programming and data: Python (NumPy, pandas, scikit-learn), C/C++, MATLAB, Simulink, data analysis, test validation

Control and automation: PLC and DCS architectures, PID control, fuzzy logic control, condition monitoring, Mahalanobis Distance health indexing

Engineering practice and CAD: Commissioning, QA and calibration records, technical reports, equipment testing, SolidWorks, Fusion 360

ENGINEERING EXPERIENCE

Student Engineer – Control and Instrumentation | Botswana Power Corporation, Morupule B Power Station | *Feb–Jul 2024* | *~6-months*

- Assisted with fault isolation, rewiring and maintenance of industrial control loops and ignitor control cabinets.
- Supported calibration and functional checks of motor-operated valves, RTDs and thermocouples.
- Participated in loop verification during plant start-up and outage activities.
- Maintained calibration records and equipment punch lists in accordance with site quality procedures.
- Reported instrument status and coordinated assigned maintenance tasks with engineers and control-room personnel.

Oxygen Plant Installation and Commissioning – Project Volunteer | Kanye Adventist Hospital | *Dec 2025–Jan 2026*

- Supported installation, electrical wiring and integration of compressors, dryer, pressure swing adsorption (PSA) equipment and oxygen booster system.
- Assisted with piping connections, supports and drainage arrangements, including oxygen storage and cylinder-filling connections.
- Participated in functional checks of pressure regulation, flow-control logic and alarm setpoints; supported successful cylinder-filling tests.

SELECTED ENGINEERING PROJECTS

OPTIBELT – Optimised Belt and Stockpile Monitoring System | BIUST | *2026* | *Team project*

- Contributed to a modular monitoring concept combining LiDAR-based stockpile volume estimation and YOLOv8-based conveyor belt wear detection.
- Developed a Mahalanobis Distance condition-health index in Python; benchmarked outputs against an earlier MATLAB implementation across thousands of test windows and calibrated load-dependent thresholds.
- Integrated condition-monitoring data on a Jetson Nano with UART communication to an ESP32 development module; applied fuzzy-PID control logic for belt speed response to fault indicators.
- Adapted the embedded implementation to an ESP32 development module after the STM32 board proved faulty.
- Prepared a technical project proposal aimed at Morupule Coal Mine; project received the programme's Best Final Year Project award.

ADDITIONAL TECHNICAL PROJECTS

Environmental Telemetry Monitoring System | BIUST, TELE 523 PBL | *2026*

- Contributed to an air-quality telemetry signal chain using 8-bit quantisation, PCM encoding, NRZ-L line coding and parity checks.
- Evaluated system behaviour under signal noise, sensor drift and data loss.

Conveyor Pullcord Fault Detection – Proposed Design | BPC Control and Instrumentation | *2025*

- Designed a proposed conveyor emergency-stop pullcord monitoring circuit to distinguish open-circuit, short-circuit and intermittent-contact faults.

EDUCATION

Bachelor of Engineering – Mechatronics and Industrial Instrumentation | BIUST, Palapye | *2021–2026*

Graduated 11 September 2026 | GPA: 3.79 | Best Final Year Project (team award)

AWARDS AND ACHIEVEMENTS

- Best Final Year Project, BEng Mechatronics and Industrial Instrumentation – BIUST (OPTIBELT, 2026).
- First Place, Botho Hacks 2024 – led a team developing an AI computer-vision wildlife detection solution.
- Outstanding Cambridge Learner Award, IGCSE Design and Technology – Botswana (November 2020).

LEADERSHIP AND TECHNICAL VOLUNTEERING

- Teaching Assistant, BIUST / IEEE Student Branch – supported practical sessions in microcontroller programming and circuit integration.
- Organising Team, Deep Learning IndabaX Botswana – supported delivery of a regional machine-learning event and hackathon.
- Secretary, Institute of Robotics, Instrumentation and Hardware (IRIH) – managed communications and helped organise technical workshops.
- President, BIUST Adventist Fellowship Club – coordinated student activities and community outreach.

ADDITIONAL EXPERIENCE

Marine and Freshwater Fish Handler | Hyde Aqua (Pty) Ltd, Botswana | *Feb–Jul 2021*

- Provided customer guidance, monitored aquatic system health and supported quarantine and care procedures.

EARLIER EDUCATION

International General Certificate of Secondary Education (IGCSE) | Eastern Gate Academy, Francistown | 2017–2020